

VRATZA, Bulgaria

WMO: 155050

Lat: 43.256N

Lon: 23.533E

Elev: 1017

StdP: 14.16

Time Zone: 2.00 (EUE)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.1	16.2	6.6	8.0	12.2	11.8	10.4	17.9	32.0	45.2	26.6	45.3	2.8	140	0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	92.6	70.9	89.4	69.9	86.3	68.9	73.4	87.9	71.6	85.7	70.1	83.7	5.3	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.4	108.4	81.9	66.8	102.5	79.5	65.4	97.3	77.2	37.6	87.9	36.0	85.9	34.7	83.9	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
23.7	16.6	13.1	DB	6.0	98.3	6.0	4.7	1.7	101.7	-1.8	104.5	-5.2	107.1	-9.5	110.5
			WB	5.4	77.2	6.0	2.9	1.1	79.3	-2.4	81.0	-5.8	82.6	-10.1	84.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.0	32.9	36.6	45.0	53.9	63.1	69.6	73.9	73.6	64.9	54.2	44.3	34.6	(6)
	DBStd	16.60	9.92	10.80	9.60	7.05	6.54	6.21	5.69	5.80	6.79	8.14	8.57	9.56	(7)
	HDD50	1928	534	387	214	39	1	0	0	0	0	51	216	486	(8)
	HDD65	4891	995	796	620	338	114	26	4	6	83	344	623	942	(9)
	CDD50	3382	4	11	60	157	407	588	742	732	446	182	44	9	(10)
	CDD65	870	0	0	1	6	55	164	280	274	79	10	1	0	(11)
	CDH74	7615	0	0	8	51	394	1382	2640	2470	597	72	1	0	(12)
	CDH80	2771	0	0	0	8	69	452	1068	994	172	8	0	0	(13)
(14) Wind	WSAvg	4.7	4.4	5.1	6.1	5.6	4.8	4.8	4.9	4.5	4.3	3.7	4.3	4.3	(14)
(15) Precipitation	PrecAvg	29.7	1.7	1.6	2.2	2.9	3.7	3.9	3.0	2.2	2.2	2.0	2.3	2.1	(15)
	PrecMax	43.4	5.5	3.6	5.8	6.9	11.2	9.2	7.2	6.8	7.3	6.1	5.6	4.6	(16)
	PrecMin	15.6	0.0	0.1	0.7	0.3	0.5	0.9	0.4	0.0	0.2	0.0	0.4	0.1	(17)
	PrecStd	5.6	1.2	0.8	1.1	1.5	2.0	2.0	1.8	1.8	1.9	1.4	1.3	1.2	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.5	65.3	73.6	80.6	85.7	92.7	99.8	97.0	90.3	80.7	71.4	65.2	(19)
		MCWB	48.8	51.0	56.3	61.1	67.2	71.9	70.2	72.5	68.6	64.4	57.5	53.4	(20)
	2%	DB	55.4	60.1	68.4	73.3	81.5	88.5	93.3	92.4	84.8	75.3	66.6	59.5	(21)
		MCWB	46.3	48.8	53.3	58.2	65.8	70.8	70.9	70.7	66.4	62.1	55.6	49.3	(22)
	5%	DB	50.8	56.6	64.2	69.1	78.3	85.0	89.4	89.1	80.5	70.9	61.4	53.6	(23)
		MCWB	43.6	46.9	51.5	56.0	64.3	69.3	69.7	69.8	64.4	60.1	52.4	45.9	(24)
	10%	DB	46.9	52.8	60.3	65.8	75.2	81.9	86.0	85.9	76.6	66.5	56.6	47.7	(25)
		MCWB	41.1	44.8	49.4	54.4	62.6	67.9	68.8	68.5	63.0	57.8	50.4	42.3	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.0	52.1	58.0	63.4	69.6	75.0	75.4	75.6	70.5	66.1	58.6	54.4	(27)
		MCDB	59.0	62.9	71.2	76.6	82.8	87.6	90.7	91.1	85.3	78.6	69.3	64.0	(28)
	2%	WB	46.7	49.5	54.1	59.3	67.1	72.6	72.9	73.3	67.4	63.3	56.0	49.4	(29)
		MCDB	54.6	59.3	65.7	70.5	79.0	85.6	87.8	88.5	81.5	73.0	64.8	58.4	(30)
	5%	WB	44.0	47.3	52.0	57.2	65.3	70.5	71.3	71.3	65.6	60.7	53.5	46.2	(31)
		MCDB	50.2	55.7	63.9	67.2	76.6	83.0	86.1	85.1	78.4	69.0	59.8	53.5	(32)
	10%	WB	41.3	44.9	49.7	55.3	63.7	68.6	69.8	69.5	63.8	58.4	50.9	43.1	(33)
		MCDB	46.1	52.1	59.7	63.9	73.8	80.2	83.7	83.1	74.4	65.6	56.1	47.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	11.0	12.8	14.5	15.4	16.8	17.3	18.5	18.7	16.7	14.4	12.0	10.2	(35)
		MCDBR	18.4	19.4	20.0	20.7	21.1	22.1	23.9	23.2	22.8	21.4	20.3	18.3	(36)
		MCWBR	11.8	12.0	11.0	11.2	10.9	10.1	10.0	9.7	10.4	11.2	11.5	11.2	(37)
	5% WB	MCDBR	17.7	18.8	19.6	18.3	19.8	20.5	21.3	21.2	20.8	19.8	18.6	17.2	(38)
		MCWBR	11.5	12.0	11.4	10.7	11.0	10.7	9.9	10.2	10.3	11.1	11.0	10.9	(39)
(40) Clear-Sky Solar Irradiance	taub		0.328	0.356	0.398	0.456	0.441	0.438	0.446	0.458	0.429	0.386	0.352	0.329	(40)
	taud		2.339	2.266	2.205	2.054	2.166	2.213	2.193	2.157	2.212	2.322	2.382	2.369	(41)
	Ebn at Noon		251	262	265	257	265	266	262	254	252	248	240	239	(42)
	Edh at Noon		26	33	40	50	46	44	45	45	40	31	25	23	(43)
(44) All-Sky Solar Radiation	RadAvg		476	737	1076	1451	1757	1973	2024	1813	1319	871	518	395	(44)
	RadStd		72	98	119	169	154	153	137	149	105	114	66	56	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	+1.26	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)	+0.90	N/A	N/A	N/A	N/A	+0.74	N/A	N/A	-239	N/A	N/A

Nomenclature: See separate page